Data Cleaning Process

July 5, 2026

1 Data Cleaning Process

2 Grocery Inventory & Sales Performance Dashboard

2.1 Microsoft Excel Data Analytics Project

Project Name: Grocery Inventory & Sales Performance Dashboard

Domain: Grocery Retail & Inventory Management

Tool Used: Microsoft Excel

Project Type: Data Analytics Portfolio Project

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4 1. Introduction

Data cleaning is one of the most important stages in the data analytics lifecycle. Before creating dashboards or performing business analysis, raw data must be cleaned, standardized, and validated to ensure accurate reporting.

The grocery inventory dataset contained inconsistent date formats, currency values stored as text, and missing business metrics required for analysis. Therefore, a structured data cleaning process was carried out before developing the dashboard.

The cleaned dataset became the foundation for all Pivot Tables, Pivot Charts, KPI calculations, and interactive dashboard components.

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5 2. Dataset Overview

The dataset contains inventory and sales information for grocery products.

5.0.1 Dataset Summary

Item	Details
Domain	Grocery Retail
Total Records	990 Products
Analysis Period	2024–2025
Tool Used	Microsoft Excel

5.0.2 Original Dataset Fields

- Product_ID
- Product_Name
- Category
- Supplier_ID
- Supplier_Name
- Stock_Quantity
- Reorder_Level
- Reorder_Quantity
- Unit_Price
- Sales_Volume
- Inventory_Turnover_Rate
- Date_Received
- Last_Order_Date
- Expiration_Date

The dataset contained operational information required to analyze inventory and sales performance.

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6 3. Data Quality Assessment

Before analysis, the dataset was reviewed to identify quality issues.

6.0.1 Issues Identified

Issue	Impact
Mixed Date Formats	Timeline and Pivot Table grouping failed
Currency stored as Text	Revenue calculations were not possible
Revenue Column Missing	Required calculated field
Inventory Value Missing	Required KPI calculation
Reorder Status Missing	Required dashboard analysis
Inconsistent Data Types	Could affect Pivot Tables and formulas

These issues were resolved before creating the dashboard.

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7 4. Data Cleaning Objectives

The primary objectives of data cleaning were:

- Standardize all date formats.
- Convert currency values into numeric format.
- Create missing business metrics.
- Improve data consistency.
- Prepare the dataset for Pivot Tables.
- Enable Timeline filtering.
- Ensure accurate KPI calculations.
- Build a reliable dataset for dashboard development.

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8 5. Data Cleaning Process

8.1 Step 1 – Inspect the Dataset

The first step involved reviewing every column to understand the dataset structure and identify potential problems.

The inspection focused on:

- Missing business metrics
- Incorrect data types
- Mixed date formats
- Currency stored as text
- Formula requirements
- Dashboard requirements

This helped determine the transformations required before analysis.

8.2 Step 2 – Standardize Date Formats

8.2.1 Problem

The dataset contained inconsistent date formats.

Examples:

Original Value	Format
8/16/2024	MM/DD/YYYY
3/18/2024	MM/DD/YYYY
11-01-2024	DD-MM-YYYY
08-03-2024	DD-MM-YYYY

Some values were stored as Excel dates while others were stored as text.

This caused:

- Incorrect sorting
- Timeline filter issues
- Pivot Table grouping problems

8.2.2 Solution

A helper column was created to convert every date into a valid Excel Date.

8.2.3 Formula Used

```
=IF(
ISNUMBER(K2),
K2,
DATE(
RIGHT(K2,4),
LEFT(K2,FIND("/",K2)-1),
MID(K2,FIND("/",K2)+1,FIND("/",K2,FIND("/",K2)+1)-FIND("/",K2)-1)
)
)
```

The same logic was applied to:

- Cleaned_Date_Received
- Cleaned_Last_Order_Date
- Cleaned_Expiration_Date

Finally, all cleaned dates were formatted as:

```
DD-MM-YYYY
```

8.2.4 Formula Explanation

The formula performs the following steps:

Step 1

Checks whether the value is already a valid Excel date.

`ISNUMBER(K2)`

If TRUE, the original date is returned.

Step 2

Extract the Year.

`RIGHT(K2,4)`

Example:

```
8/16/2024

↓

2024
```

Step 3

Extract the Month.

`LEFT(K2,FIND("/",K2)-1)`

Example:

```
8/16/2024

↓

8
```

Step 4

Extract the Day.

`MID(K2,FIND("/",K2)+1,FIND("/",K2,FIND("/",K2)+1)-FIND("/",K2)-1)`

Example:

```
8/16/2024

↓

16
```

Step 5

Rebuild the Excel Date.

`DATE(2024,8,16)`

Excel converts this into a valid date serial number.

After formatting:

`16-08-2024`

8.2.5 Business Benefit

Standardizing dates enabled:

- Correct sorting
- Timeline functionality
- Monthly grouping
- Pivot Table analysis
- Monthly Revenue Trend chart

8.3 Step 3 – Convert Unit Price

8.3.1 Problem

The Unit_Price column contained currency values stored as text.

Examples:

`$4.50`

`$18.00`

`$22.75`

These values could not be used in mathematical calculations.

8.3.2 Solution

The dollar symbol was removed and values converted into numbers.

8.3.3 Formula

`=VALUE(SUBSTITUTE([@Unit_Price],"$",""))`

The cleaned values were formatted as Currency (USD).

8.4 Step 4 – Create Revenue

The original dataset did not include Revenue.

Revenue was calculated using:

`Revenue = Sales Volume × Unit Price`

8.4.1 Formula

`=[@Sales_Volume]*[@Cleaned_Unit_Price]`

Revenue became the primary metric for sales analysis.

8.5 Step 5 – Create Inventory Value

Inventory Value measures the monetary value of products currently available in stock.

Business Formula

`Inventory Value = Stock Quantity × Unit Price`

8.5.1 Excel Formula

`=[@Stock_Quantity]*[@Cleaned_Unit_Price]`

This metric was used in KPI cards and category-level inventory analysis.

8.6 Step 6 – Create Reorder Status

The dataset did not contain a Reorder Status column.

A business rule was implemented.

8.6.1 Business Logic

```
If

Stock Quantity < Reorder Level

↓

Reorder Required

Otherwise

↓

Sufficient Stock
```

8.6.2 Excel Formula

```
=IF([@Stock_Quantity]<[@Reorder_Level],"Reorder Required","Sufficient Stock")
```

This field was used for KPI calculations, slicers, and the Reorder Status Distribution chart.

8.7 Step 7 – Create Month_Year

To build the Monthly Revenue Trend chart, a Month_Year helper column was created.

8.7.1 Formula

```
=DATE(YEAR([@Cleaned_Date_Received]),MONTH([@Cleaned_Date_Received]),1)
```

8.7.2 Purpose

- Monthly grouping
- Timeline filtering
- Monthly Revenue Trend
- Pivot Table grouping

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9 6. Calculated Columns

The following calculated columns were created during data preparation.

Column	Purpose
Cleaned_Date_Received	Standardized dates
Cleaned_Last_Order_Date	Standardized dates
Cleaned_Expiration_Date	Standardized dates
Cleaned_Unit_Price	Numeric currency
Revenue	Sales analysis
Inventory_Value	Inventory analysis
Reorder_Status	Stock monitoring
Month_Year	Monthly trend analysis

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10 7. Data Validation

The cleaned dataset was validated before dashboard development.

Validation checks included:

- Date verification
- Numeric data verification
- Formula consistency

- Revenue calculation
- Inventory Value calculation
- Reorder logic
- Pivot Table verification
- Dashboard filter testing

All calculations were verified before creating the dashboard.

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11 8. Challenges Faced

Several practical challenges were encountered during the project.

11.0.1 Challenge 1

Mixed date formats.

Solution

Standardized all dates using Excel functions.

11.0.2 Challenge 2

Currency stored as text.

Solution

Converted Unit Price into numeric values.

11.0.3 Challenge 3

Missing business metrics.

Solution

Created Revenue and Inventory Value columns.

11.0.4 Challenge 4

No Reorder Status.

Solution

Implemented business logic using IF().

11.0.5 Challenge 5

Monthly Trend Analysis.

Solution

Created Month_Year helper column to support Timeline filtering and monthly grouping.

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12 9. Final Cleaned Dataset

The final dataset contained both original and calculated fields.

12.0.1 Original Fields

- Product_ID
- Product_Name
- Category
- Supplier_ID
- Supplier_Name
- Stock_Quantity
- Reorder_Level
- Reorder_Quantity
- Sales_Volume
- Inventory_Turnover_Rate

12.0.2 Calculated Fields

- Cleaned_Date_Received
- Cleaned_Last_Order_Date
- Cleaned_Expiration_Date
- Cleaned_Unit_Price
- Revenue
- Inventory_Value
- Reorder_Status
- Month_Year

This dataset became the primary source for all dashboard components.

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13 10. Business Benefits

The data cleaning process provided several business benefits.

- Improved data quality.
- Accurate KPI calculations.
- Reliable Pivot Tables.
- Interactive Timeline filtering.
- Consistent monthly analysis.

- Accurate dashboard reporting.
- Better business decision-making.

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14 11. Conclusion

The Data Cleaning Process transformed raw grocery inventory data into a structured, analysis-ready dataset suitable for business reporting.

Through standardizing dates, converting currency values, creating calculated columns, validating data types, and implementing business rules, the dataset became reliable for KPI development, Pivot Tables, Pivot Charts, and dashboard creation.

This process demonstrates practical skills in **data cleaning, transformation, validation, and business analytics**, which are essential responsibilities of a Data Analyst.